Methoxyflurane

Mechanism of action

- Methoxyflurane is an inhalational analgesic.
- The mechanism of action is not clear.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Acute moderate to severe pain if:
  - An opiate cannot be administered because of a contraindication or caution, or
  - Opiate administration will be significantly delayed, or
  - There will be severe worsening of pain during a procedure or extrication.

Contraindications

- ✕ Known severe allergy.
- ✕ Personal or family history of malignant hyperthermia.
- ✕ Known renal impairment.
- ✕ 15 ml methoxyflurane (5 doses) administered within the last week. Frequent administration increases the risk of renal impairment.

Cautions

- ⚠ Age greater than or equal to 75 years, particularly if frail.
- ⚠ Pre-eclampsia. In this setting, renal impairment is likely and methoxyflurane may worsen renal impairment.
- ⚠ Administration within a confined space.
- ⚠ Acute exacerbation of chronic pain.

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Use in pregnancy or when breastfeeding

- Safety has not been formally demonstrated in pregnancy, but methoxyflurane may be administered. Methoxyflurane has been extensively used during labour in Australia and New Zealand for many years without adverse effects.
- Methoxyflurane may cause temporary drowsiness in the baby and administration should be discussed with the lead maternity carer if there are known signs of foetal distress.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Maximum of 6 ml (two doses) for a patient aged greater than or equal to 12 years.
- Maximum of 3 ml (one dose) for a child aged less than 12 years.

Administration

- Whenever possible, have the patient self-administer methoxyflurane.
- Administer 3 ml (one dose) at a time and always use the charcoal filter. Using a charcoal filter maximises the amount of exhaled methoxyflurane that is absorbed, limiting exposure to personnel.
- Instruct the patient to breathe out through the inhaler.
- Do not administer supplementary oxygen via the inhaler as this significantly increases the amount of methoxyflurane lost through evaporation.
- Place the inhaler in a closed plastic bag if the methoxyflurane has not been fully used. It may subsequently be reused by the same patient.
- An ambulance is not considered a confined space. Maximising ventilation (for example, having ventilation fans on) reduces occupational exposure. Consider not administering methoxyflurane in an ambulance if the patient cannot cooperate with breathing out through the inhaler.

Common adverse effects

- Sedation.
- Feeling light-headed.

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Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 2-5 minutes after stopping administration.

Usual preparation

- 3 ml bottle accompanying a plastic inhaler.

Pharmacokinetics

- Approximately 20% is exhaled. The remainder is metabolised in the liver.
- One of the metabolites is fluoride ions. High concentrations of fluoride ions have been associated with renal impairment and this is the reason for known renal impairment being a contraindication and for having a maximum dose.

Common interactions

- The effects will be increased in the presence of other pain relieving medicines or sedatives, for example opiates, benzodiazepines or alcohol.

Additional information

- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including methoxyflurane), the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Renal failure with dialysis is not a contraindication or a caution to methoxyflurane administration because once a patient is receiving dialysis further renal impairment is of no clinical consequence.
- Kidney stones and/or renal colic are not a contraindication or a caution to methoxyflurane administration because these are rarely associated with renal impairment.
- Methoxyflurane must not be left with a patient who is not being transported to a medical facility by ambulance.

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More Information

Version history

16 June 2026 1.1.0.1

Update: Indications updated.

11 September 2023 1.0.4

Update: A new caution has been added (acute exacerbation of chronic pain).

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia CPG.